CHAPTER 1 • GRADE 4 • WEEK 1

# Energy Causes Change

DISCOVERY • EVIDENCE

**Where can we observe evidence of energy causing change?**

ENERGY → CHANGE → ?



# Big Question

BIG QUESTION

## Where can we observe evidence of energy causing change?

Keep it in mind all chapter. We will come back to it.

**Think: What is ONE change you have seen happen?**

ENERGY → CHANGE → ?

# Big Question — My first thinking

*Guided Notes p. 1*

## Where can we observe evidence of energy causing change?

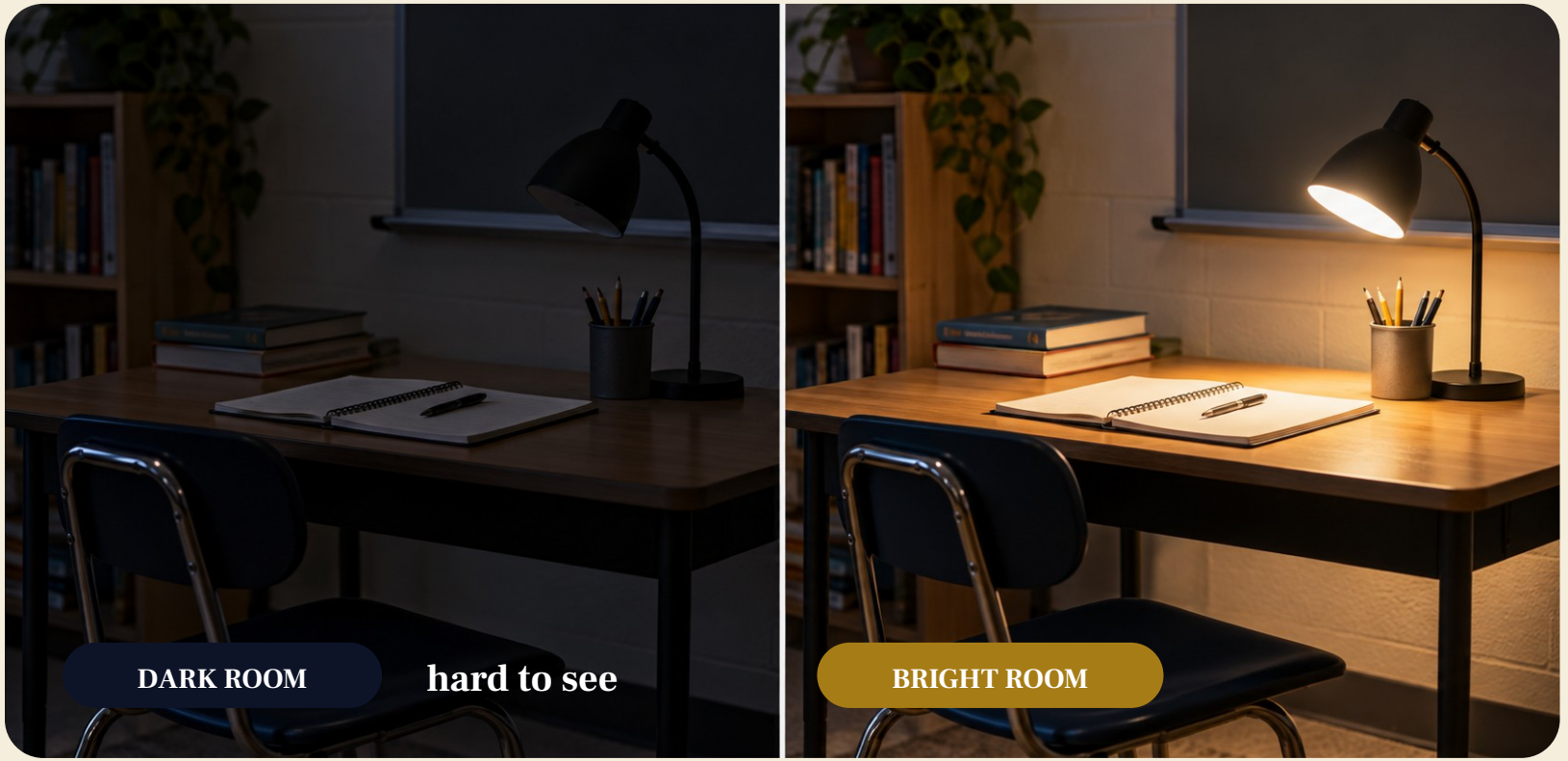
### My first thinking

*Write 1–2 quick ideas. No wrong answer yet.*

OBSERVE

# What changed?

*Reader p. 1 — you walk into a dark room and flip a switch.*




**Something became different.**

**That is a change.**

light fills the room

TURN & TALK

# I notice…  I wonder…



I NOTICE

**The room is different now.**
**Light is everywhere.**



I WONDER

**What made the light change so fast?**

**Share one notice and one wonder with a partner.**

OBSERVE

# Electric Pencil Sharpener



BEFORE

## Observe the pencil BEFORE we sharpen it.

**A dull pencil — tip is flat, and it is quiet.**

*Look closely. What do you see? What do you hear?*

NEXT: SHARPENER DEMO

PREDICT

# Predict: what will change?

*Think before we turn it on.*

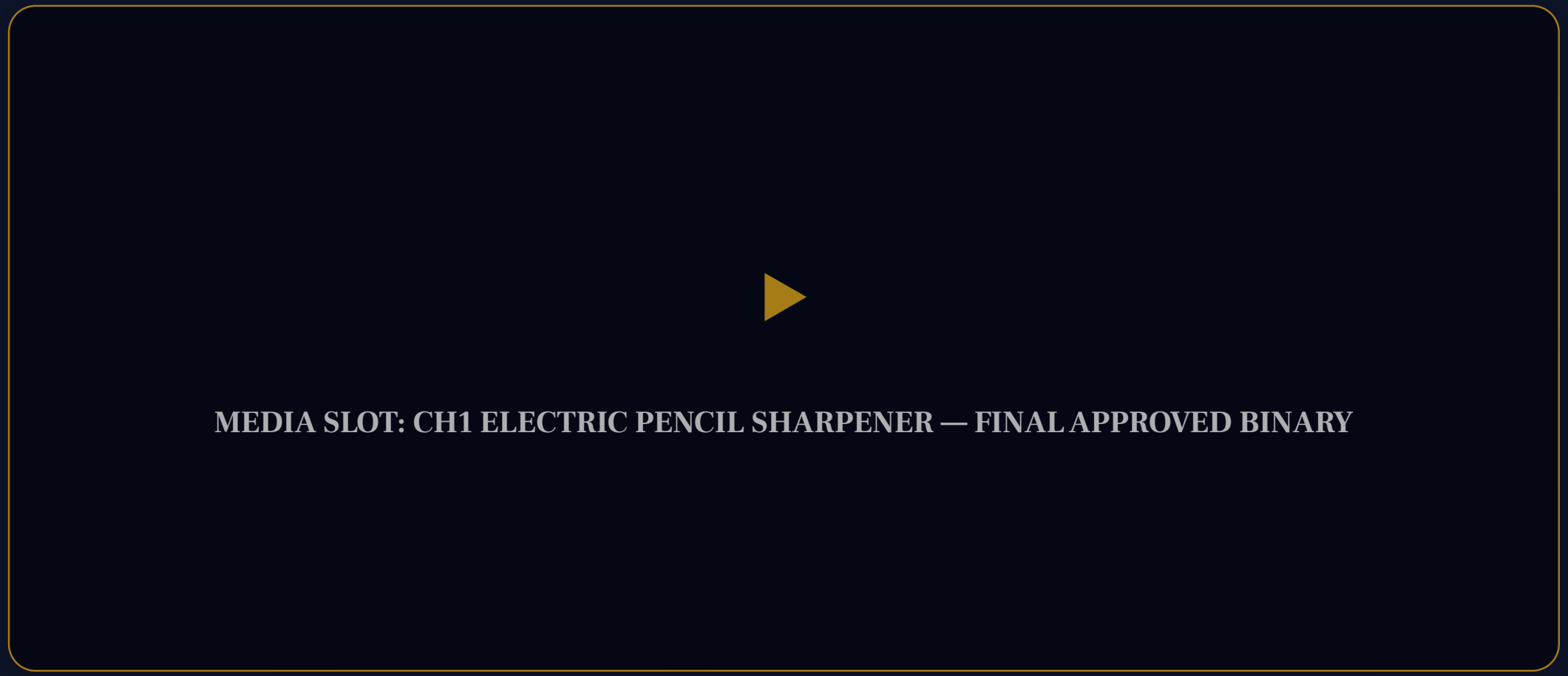
## When the sharpener runs, the pencil will…

**get** _____

**Make a prediction. Say it to your partner.**

WATCH CLOSELY

# Electric Pencil Sharpener Demo




## What changes do you see or hear?

If the video is unavailable, use the live demo or continue to the evidence slide.

OBSERVE

# What changed?







**CHANGES**

- the tip is sharper
- the pencil is shorter
- the sharpener makes a sound

**The changes are clues that energy was involved.**

# Vocabulary

KEY WORD

## energy



**the ability to cause change**

Reader p. 1

RECITE

# Vocabulary

KEY WORD

## change



to become different

Reader p. 1

# Vocabulary — Energy & Change

*Guided Notes • Reader p. 1*

**energy — the ability to cause _____ .**

**change — to become _____ .**

*Write the missing word in each blank.*

# Vocabulary — Energy & Change

*Guided Notes • Reader p. 1*

**energy — the ability to cause change .**

**change — to become different .**

**change • different**

OBSERVE

# Look at the seashore

*Reader p. 2 — the seashore changes all around us.*

**Waves move.**
**Sand shifts.**
**Water makes sound.**

ENERGY → CHANGE → ?

**What are these changes a clue about?**



# N1 — Energy Causes Change

*Guided Notes • Reader pp. 1–2*




**Energy has the ability to cause change.**

**When you see a change, you are seeing _____ of energy.**

*Write the missing word. Leave the last box of your model empty for now.*

# N1 — Energy Causes Change

*Guided Notes • Reader pp. 1–2*

**ENERGY** → **CHANGE** → **EVIDENCE**

**Energy has the ability to cause change.**

**When you see a change, you are seeing evidence of energy.**

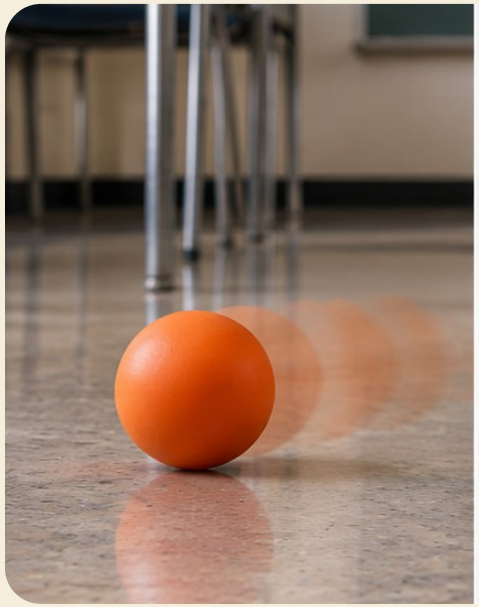
**Now write the same word in the last box of your model.**

OPTIONAL MEDIA: ENERGY DETECTIVE

RECITE

# Energy has different forms

*Where can we observe evidence of energy causing change?*



MOTION



SOUND



LIGHT



HEAT



ELECTRICAL ENERGY

**Say the five forms together.**

OBSERVE

# Five forms of energy

MOTION

SOUND

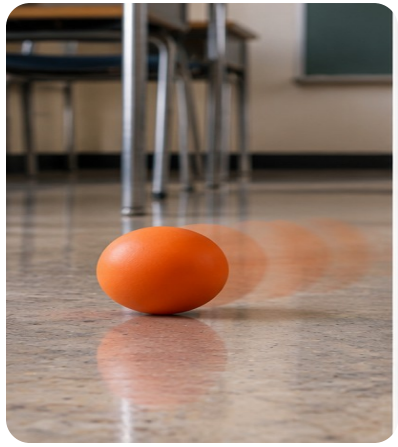
LIGHT

HEAT

ELECTRICAL ENERGY

OBSERVE

# Energy has different forms



## motion

A rolling ball moves.



## sound

A bell makes sound.



## light

A lamp makes an area brighter.



## heat

A warm mug gives off heat.



## electrical energy

A fan uses electrical energy to operate.

# N2 — Energy Occurs in Different Forms

*Guided Notes • Reader p. 3*

*Write one example that shows a change for each form.*

| FORM | ONE EXAMPLE THAT SHOWS A CHANGE |
|---|---|
| **motion** | |
| **sound** | |
| **light** | |
| **heat** | |
| **electrical energy** | |

# N2 — Energy Occurs in Different Forms

*Guided Notes • Reader p. 3*

SAMPLE ANSWERS — YOUR EXAMPLES MAY DIFFER

| FORM | ONE EXAMPLE THAT SHOWS A CHANGE |
|---|---|
| motion | a ball moves across the floor |
| sound | a bell rings |
| light | a dark area becomes brighter |
| heat | something becomes warmer |
| electrical energy | a fan begins moving |

# Vocabulary

KEY WORD

## stored energy



**energy that has the ability to cause change at a later time**

Reader p. 4

# Vocabulary — Stored Energy

*Guided Notes • Reader p. 4*

**stored energy — energy that has the ability to cause change at a _____ time.**

*Write the missing word in the blank.*

# Vocabulary — Stored Energy

*Guided Notes • Reader p. 4*

**stored energy — energy that has the ability to cause change at a later time.**

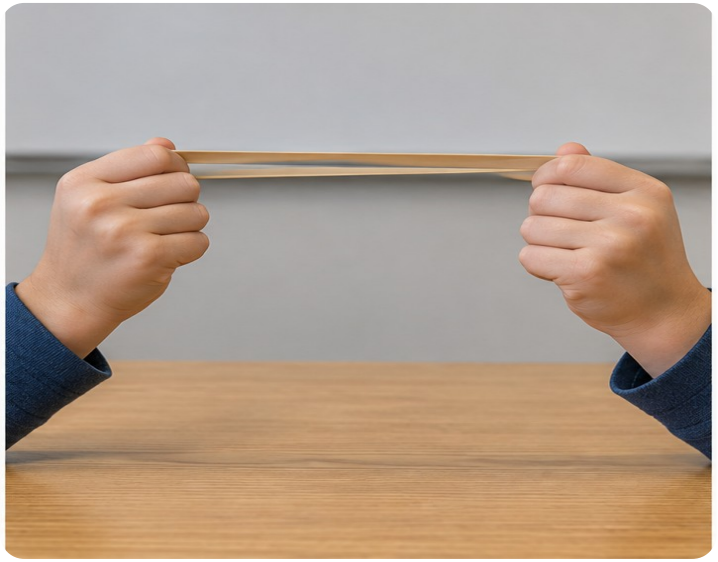
**later**

OBSERVE

# Objects can store energy



battery (in a flashlight)



stretched rubber band



raised object (apple)

## What change could happen later?

# N3 — Objects Can Store Energy

*Guided Notes • Reader pp. 4–5*

**STORED ENERGY** → **RELEASE** → **CHANGE**

| EXAMPLE | WHAT CHANGE COULD HAPPEN LATER? |
|---|---|
| battery (in a flashlight) | |
| stretched rubber band | |
| raised object (apple) | |

**Objects also have stored energy because of their _____ .**

# N3 — Objects Can Store Energy

*Guided Notes • Reader pp. 4–5*

**STORED ENERGY** → **RELEASE** → **CHANGE**

| EXAMPLE | WHAT CHANGE COULD HAPPEN LATER? |
|---|---|
| battery (in a flashlight) | the flashlight lights up |
| stretched rubber band | it snaps and makes a sound |
| raised object (apple) | it falls down |

**Objects also have stored energy because of their position .**

position • Sample answers shown — yours may differ.

# Recognizing Evidence

## What changed? What is the evidence?



LIGHT



PENCIL



MOTION

*Describe only what you can observe.*

# N4 — Recognizing Evidence

*Guided Notes • Reader pp. 1–5*

*Draw a change you observed. Show what it looked like BEFORE and AFTER.*

BEFORE



AFTER

## What changed?

# Check Your Thinking

*Guided Notes • Reader pp. 1–5*

**How do we know energy is involved if we cannot see energy itself?**

**Start with: “We know because we see …”**

# Check Your Thinking

*Guided Notes • Reader pp. 1–5*

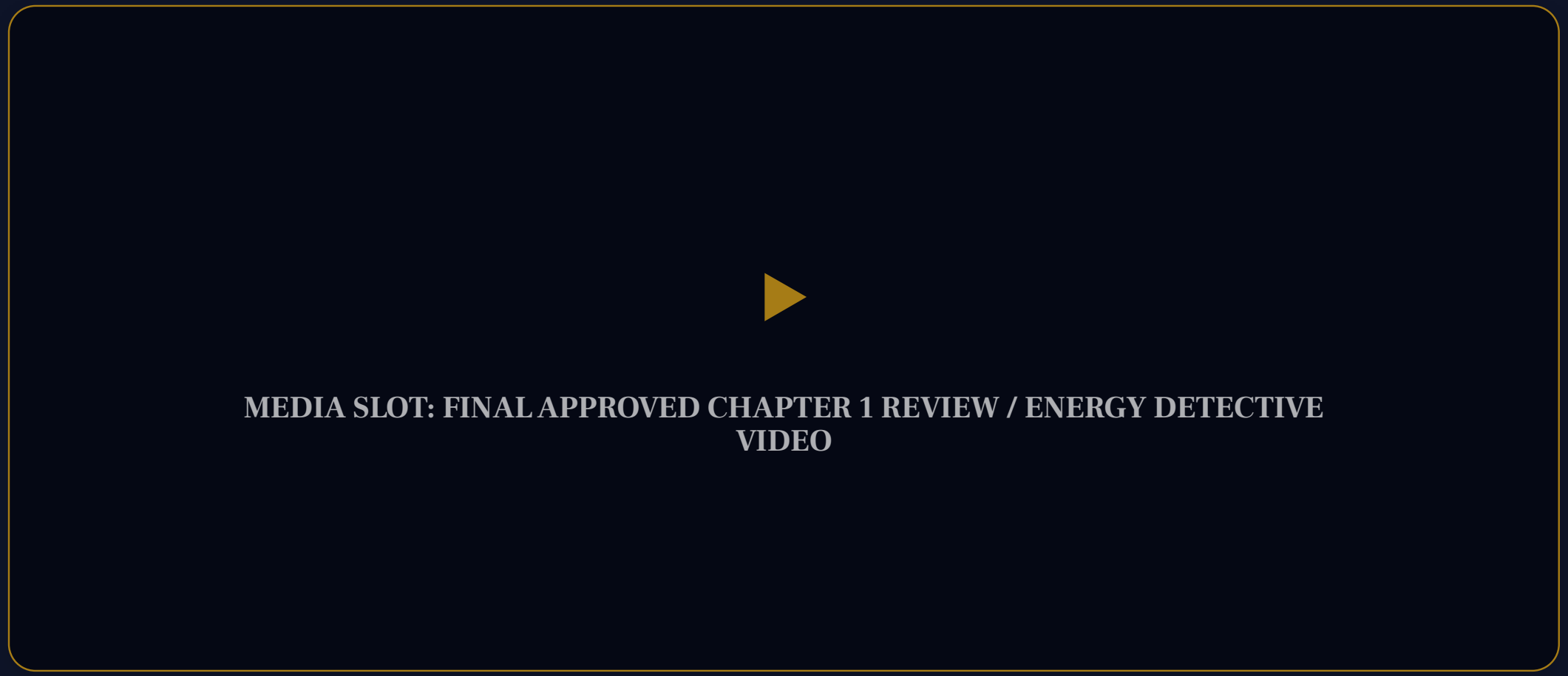
SAMPLE

**“We know because we see a change, like a ball starting to move or a light turning on.”**

**A change you observe is evidence of energy.**

OPTIONAL • SKIP IF SHORT ON TIME

# CHAPTER 1 OPTIONAL REVIEW




**Listen for energy, change, and evidence.**

# The Big Question Revisited

## Where can we observe evidence of energy causing change?

**MY FIRST THINKING**

*Look back at page 1.*

**NOW I THINK**

ENERGY → CHANGE → EVIDENCE

VOTE

# Review 1 — concept

## Energy is best described as —

A a visible glow

B something only batteries have

C the ability to cause change

D a kind of fuel

CHECK

# Review 1 — answer

## Energy is best described as —

A a visible glow

B something only batteries have

✓ C **the ability to cause change**

D a kind of fuel

**Energy is the ability to cause change (Reader p. 1).**

VOTE

# Review 2 — visual

## A pencil is dull, then you sharpen it. What change do you observe?



A It gets shorter and sharper

B It gets longer

C It changes color

D It disappears

CHECK

# Review 2 — answer

## A pencil is dull, then you sharpen it. What change do you observe?



✓ A **It gets shorter and sharper**

B It gets longer

C It changes color

D It disappears

**The pencil becomes shorter and sharper — a change.**

VOTE

# Review 3 — scenario

## Which object has stored energy that can cause a change LATER?



A. A ball already rolling

B. A toy already moving

C. A flashlight already on

D. A stretched rubber band

# Review 3 — answer

## Which object has stored energy that can cause a change LATER?



A — A ball already rolling

B — A toy already moving

C — A flashlight already on

✓ D — **A stretched rubber band**

**The rubber band stores energy; it changes when released.**

VOTE

# Review 4 — comparison

## A carousel spins and plays music. Which two forms of energy come from the spinning and the music?

**A** only light

**B** motion and sound

**C** motion and silence

**D** sound and stillness

CHECK

# Review 4 — answer

## A carousel spins and plays music. Which two forms of energy come from the spinning and the music?

A only light

✓ B **motion and sound**

C motion and silence

D sound and stillness

**The spinning is motion; the music is sound (Reader p. 3).**

VOTE

# Review 5 — concept

## How do we know energy is involved if we cannot see it?

**A** We observe a change happen

**B** Energy can always be seen

**C** We guess without evidence

**D** Nothing can change

# Review 5 — answer

## How do we know energy is involved if we cannot see it?

✓ A **We observe a change happen**

B Energy can always be seen

C We guess without evidence

D Nothing can change

**We see a change — like movement — as evidence.**

VOTE

# Review 6 — scenario

## A battery sits still in a flashlight. Does it have energy?



- **A** No — still objects have no energy
- **B** Only when it is moving
- **C** Yes — stored energy that can cause change later
- **D** Only when the flashlight is on

CHECK

## A battery sits still in a flashlight. Does it have energy?

✓ C

**Yes — stored energy that can cause change later**

A battery stores energy. It can cause a change later.

ENERGY → CHANGE → EVIDENCE

**Look for the change. Use the change as evidence that energy was involved.**

CHAPTER 1 COMPLETE